

STAT 511, SECTION 001

INTRODUCTION TO STATISTICAL THEORY

Winter 2026 Syllabus

Time: MoWe 11:30 am-1:00 pm
Classroom: 1755 SPH1
Instructor: Tailen Hsing (thsing@umich.edu)
GSIs: Noah Kochanski (noahkoch@umich.edu) and Yue Yu (yueyu@umich.edu)

Attending Classes: Our course will meet in person only.

Prerequisites: STATS 510

Course Description: This is a graduate-level introductory course to key concepts, methods and theory in statistical inference. The topics covered will include univariate and multivariate families of distributions, likelihood principle, point estimation, confidence regions, hypothesis tests, large sample properties, and other selected topics in contemporary methods.

Course Goals and Lecture Style

Beyond mastering the topics outlined in the course description, a larger goal is for you to develop proficiency in formulating solutions to statistical problems in applications. The lectures will be fast paced. So, it's important to arrive prepared, including having previewed the posted lecture notes and relevant textbook sections beforehand. Active participation will enrich your learning experience, and your questions and contributions to discussion are strongly encouraged.

Course Website: Canvas is our course management system. Course information including office hours, grades, lecture notes, and homework assignments will be posted there.

Textbook: *Statistical Inference*, second edition, by George Casella and Roger L. Berger, Duxbury.

Homework

- A tentative homework schedule is as follows:

	Assignment date	Due date
1	Jan 7	Jan 14
2	Jan 14	Jan 21
3	Jan 21	Jan 28
4	Jan 28	Feb 4
5	Feb 4	Feb 11
6	Feb 16	Feb 25
7	Feb 25	Mar 11
8	Mar 11	Mar 18
9	Mar 18	Mar 25
10	Apr 1	Apr 8
11	Apr 8	Apr 15

- Submit your work to Gradescope (accessible from canvas). Go to [Get Started](#) on the Gradescope website and watch [For Students: Submitting Homework](#). Completing your work in LaTeX is recommended but not required.
- The lowest homework grade will be dropped. **Late homework will NOT be accepted except under highly special circumstances.**

- Solutions to homework assignments will be posted on Canvas after the assignment closes.

Exams

Two in-class exams and a final exam will be given.

- **Exam 1**
Date: Wednesday, February 18
Coverage: Material covered through February 11
Format: In class and closed book
- **Exam 2**
Date: Monday, March 30
Coverage: Material covered through March 23
Format: In class and closed book
- **Final Exam**
Time and date: 4-6 pm, Tuesday, April 28
Coverage: Comprehensive
Format: In class and closed book

Attendance/participation

- Attendance/participation will be 2% of the overall grade calculation. To receive the attendance/participation credit for a lecture, you must be physically present and answer (correctly or incorrectly) at least 75% of the poll questions using iClicker Cloud. The attendance component of your grade is computed based on the formula $(\min(x, 20)/20) * 2\%$ where x = number of lectures you attend. To set up iClicker Cloud on your mobile device, please refer to <https://lsa.umich.edu/technology-services/services/learning-teaching-consulting/instructional-tools/iclicker/iclicker-cloud.html>
- Please adhere to classroom norms by arriving promptly, muting your devices, refraining from private conversations, and showing respect to others in the room, among other etiquette guidelines.

Grade

Your final course grade will be computed based on the following distribution

Participation	2%
Homework	28%
Exam 1	20%
Exam 2	20%
Final Exam	30%

Your course grade will be assigned based on the following grading scale

A+	[95,100]	B+	[80, 85)	C+	[65, 70)	D+	[50, 55)	E	Below 40
A	[90, 95)	B	[75, 80)	C	[60, 65)	D	[45, 50)		
A-	[85, 90)	B-	[70, 75)	C-	[55, 60)	D-	[40, 45)		

Academic Integrity

The academic community, like all communities, functions best when its members **treat one another with honesty, fairness, respect, and trust**. Students are expected to work and study together to foster learning and understanding of the material. However, direct copying of homework, cheating on an exam, and other conduct that violates the academic integrity and ethical standards of the LSA College community cannot be tolerated and will result in serious consequences and disciplinary action.

So, do not cheat or do not cross that line of taking collaboration too far. The penalties are steep, and the potential rewards are minor. Please see <https://lsa.umich.edu/lsa/academics/academic-integrity/academic-misconduct.html> for examples of academic misconducts. If you have a question or concern about what may be considered academic

misconduct and/or if you have any questions regarding the Stats 511 academic integrity policy, please contact your instructor.